Domain Adaptation in High-Resolution Satellite Images using Deep Learning

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1 Problem Definition

Deep Convolutional Neural Networks (DCNNs), utilizing their higher number of convolutional layers over regular CNNs, are exceptional in learning, classifying and semantically segmenting from complex, abstract representations such as Very High Resolution (VHR) Satellite Images. However, the usage of a DCNN brings forth the problem of domain shift introduced by inherent regional differences present within a class across geographical areas, resulting in performance degradation. This problem is further exacerbated by varying acquisition settings and a lack of extensive annotated datasets for each domain [2]. Moreover, the domain adaptation issue makes it particularly difficult to detect small, densely placed objects such as vehicles in satellite images [1].

2 Related Work

Recent research efforts have focused on addressing the problems of domain adaptation and generalization within satellite imagery. Research by Zhang et al. [3] and Postadjian et al. [2] emphasize how important it is to adapt DCNN models to different domains without requiring a lot of retraining. Model robustness and performance across diverse datasets have been strengthened by methods including contrastive adversarial loss, image-to-image translation, and self-supervised learning [3]. In particular, vehicle detection is addressed by Koga et al. [1], who use unsupervised domain adaptation strategies to reduce performance deterioration when the vehicle detection target region deviates from the training data domain. These approaches highlight the significance of having models that can effectively adapt to new domains, especially for applications that need accurate object detection in satellite photos.

3 Potential Solution

To overcome the aforementioned challenges, we propose a system leveraging self-supervised, domain adaption methods. Using these methods we propose an image-to-image translation model without the need for explicit domain labeling as suggested by Zhang et al [3]. By combining adversarial learning with contrastive learning, the domain gap can be greatly reduced by producing domain-adapted pictures that maintain content while

modifying their spectral properties to match the target domain. The use of unsupervised domain adaptation techniques for vehicle detection, as shown by Koga et al. [1], highlights the effectiveness of domain adaptation strategies in specific satellite imagery tasks by demonstrating the ability to maintain detection accuracy in the target domain at lower costs.

4 Conclusion

This project aims to mitigate the issues associated with domain shift, by utilizing self-supervised, domain-agnostic techniques such as image-to-image translation and contrastive adversarial learning. Since these techniques would work across a wider range of satellite imagery domains with varying acquisition settings, it would allow for a more precise analysis of VHR Satellite Images along with specific applications such as vehicle detection, aiding in the broader applicability of high-resolution satellite images for various domains.

5 Relationship to team members' background

Mayank Shetty is a Professional Master of Computer Science student, with a strong background in software development and machine learning, having worked on projects like multi-disease prediction and mood detection in audio. His experience spans the development and deployment of distributed applications over cloud-based systems. His current interest lies in the field of Computer Vision, particularly Augmented Reality, to enhance the perception of the physical environment and digital information.

Vaibhav Raheja is a Master of Autonomy and Robotics student, with a strong background in computer engineering and a track record of developing innovative solutions in robotics and machine learning. His experience spans research development in intelligent motion, hospital research internships focusing on surgical precision improvements, and leading teams in robotics competitions, showcasing his leadership and technical expertise.

Viraj Shah is a Professional Master of Computer Science student, who has a strong background in software development and machine learning models. His experience is based on developing machine learning models over cloud-based systems and has worked on projects in the field of computer vision for health care, mood detection in audio, and sales forecasting using machine learning. He is highly intrigued by the field of computer vision and how deep learning can be useful in generating improvements in the field.

References

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